

Serveur Proxy (Squid & SquidGuard)

1. Installation de Squid

Installer le service squid

```
# apt install -y squid
```

Vérifier la version

```
# squid -v
```

Vérifier le port d'écoute (par défaut 3128)

```
# ss -tlnp | grep squid
```

```
root@default:~# ss -tlnp | grep squid
LISTEN 0      256          *:3128        *.*          users:(( "squid",pid=1709,fd=12))
root@default:~#
```

Vérifier que l'utilisateur proxy existe

```
# id proxy
```

```
root@default:~# id proxy
uid=13(proxy) gid=13(proxy) groupes=13(proxy)
```

```
# groups proxy
```

```
root@default:~# groups proxy
proxy : proxy
```

2. Configuration de base

Sauvegarde du fichier de configuration

```
# cp /etc/squid/squid.conf /etc/squid/squid.conf.save
```

Supprimer les commentaires et lignes vides

```
# cat /etc/squid/squid.conf.save | grep -v ^# | grep -v ^\s$ >
/etc/squid/squid.conf
```

Le fichier devrait ressembler à ça maintenant, les commentaires sont supprimés ce qui permet une meilleure lecture du fichier.

```
GNU nano 8.4 /etc/squid/squid.conf
acl localnet src 0.0.0.1-0.255.255.255 # RFC 1122 "this" network (LAN)
acl localnet src 10.0.0.0/8 # RFC 1918 local private network (LAN)
acl localnet src 100.64.0.0/10 # RFC 6598 shared address space (CGN)
acl localnet src 169.254.0.0/16 # RFC 3927 link-local (directly plugged) machines
acl localnet src 172.16.0.0/12 # RFC 1918 local private network (LAN)
acl localnet src 192.168.0.0/16 # RFC 1918 local private network (LAN)
acl localnet src fc00::/7 # RFC 4193 local private network range
acl localnet src fe80::/10 # RFC 4291 link-local (directly plugged) machines
acl SSL_ports port 443
acl Safe_ports port 80 # http
acl Safe_ports port 21 # ftp
acl Safe_ports port 443 # https
acl Safe_ports port 70 # gopher
acl Safe_ports port 210 # wais
acl Safe_ports port 1025-65535 # unregistered ports
acl Safe_ports port 280 # http-mgmt
acl Safe_ports port 488 # gss-http
acl Safe_ports port 591 # filemaker
acl Safe_ports port 777 # multiling http
http_access deny !Safe_ports
http_access deny CONNECT !SSL_ports
http_access allow localhost manager
http_access deny manager
http_access allow localhost
http_access deny to_localhost
http_access deny to_linklocal
include /etc/squid/conf.d/*.conf
http_access deny all
http_port 3128
coredump_dir /var/spool/squid
refresh_pattern ^ftp: 1440 20% 10080
refresh_pattern -i (/cgi-bin/|\?) 0 0% 0
refresh_pattern . 0 20% 4320
```

Éditer le fichier pour ajouter les lignes suivantes à la fin :

```
# nano /etc/squid/squid.conf
```

```
cache_effective_user proxy
cache_effective_group proxy

cache_mem 16 MB
cache_dir ufs /var/spool/squid 120 16 128

cache_replacement_policy heap LFUDA
cache_swap_low 80
cache_swap_high 90
```

Redémarrer le service

```
# systemctl restart squid
```

Consulter les logs

```
# tail -f /var/log/squid/access.log
```

Elle n'affiche rien pour le moment.

Ouvrez un second terminal de la VM et fait la commande.

```
# apt install -y curl
# curl -x http://localhost:3128 http://www.google.fr
```

La commande vous renverra ceci

```
root@default:~# curl -x http://localhost:3128 http://www.google.fr
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
root@default:~# _
```

Maintenant sur le 1er terminal vous verrez les log

```
root@default:~# tail -f /var/log/squid/access.log
1772900070.321 35 :::1 TCP_MISS/301 1063 GET http://www.google.fr/ - HIER_DIRECT/172.217.2
2.99 text/html
```

3. Contrôles d'accès (ACL)

Vérifier support ACL système :

```
# cat /boot/config-$(uname -r) | grep ACL
```

```
root@default:~# tail -f /var/log/squid/access.log
1772900070.321 35 :::1 TCP_MISS/301 1063 GET http://www.google.fr/ - HIER_DIRECT/172.217.2
2.99 text/html
^C
root@default:~# cat /boot/config-$(uname -r) | grep ACL
# CONFIG_XILINX_EMACLITE is not set
CONFIG_EXT4_FS_POSIX_ACL=y
CONFIG_REISERFS_FS_POSIX_ACL=y
CONFIG_JFS_POSIX_ACL=y
CONFIG_XFS_POSIX_ACL=y
CONFIG_BTRFS_FS_POSIX_ACL=y
CONFIG_F2FS_FS_POSIX_ACL=y
CONFIG_BCACHEFS_POSIX_ACL=y
CONFIG_FS_POSIX_ACL=y
# CONFIG_NTFS3_FS_POSIX_ACL is not set
CONFIG_TMPFS_POSIX_ACL=y
CONFIG_JFFS2_FS_POSIX_ACL=y
CONFIG_ERofs_FS_POSIX_ACL=y
CONFIG_NFS_V3_ACL=y
CONFIG_NFSD_V3_ACL=y
CONFIG_NFS_ACL_SUPPORT=m
CONFIG_CEPH_FS_POSIX_ACL=y
CONFIG_9P_FS_POSIX_ACL=y
root@default:~#
```

```
# which setfacl
```

```
root@default:~# which setfacl
/usr/bin/setfacl
```

```
# which getfacl
```

```
root@default:~# which getfacl
/usr/bin/getfacl
```

Autoriser le LAN :

Ajouter dans /etc/squid/squid.conf avant "acl localnet"

```
# nano /etc/squid/squid.conf
```

A placer au débuts du fichier.

```
acl monlan src 172.17.X.0/16
```

A placer juste avant la ligne `http_access allow localhost`

```
http_access allow monlan
```

Maintenant votre fichier devrait ressembler à ça.

```
#Autoriser le LAN
acl monlan src 192.168.1.0/24

acl localnet src 0.0.0.1-0.255.255.255 # RFC 1122 "this" network (LAN)
acl localnet src 10.0.0.0/8           # RFC 1918 local private network (LAN)
acl localnet src 100.64.0.0/10        # RFC 6598 shared address space (CGN)
acl localnet src 169.254.0.0/16       # RFC 3927 link-local (directly plugged) machines
acl localnet src 172.16.0.0/12        # RFC 1918 local private network (LAN)
acl localnet src 192.168.0.0/16       # RFC 1918 local private network (LAN)
acl localnet src fc00::/7             # RFC 4193 local private network range
acl localnet src fe80::/10           # RFC 4291 link-local (directly plugged) machines

acl SSL_ports port 443
acl Safe_ports port 80               # http
acl Safe_ports port 21               # ftp
acl Safe_ports port 443              # https
acl Safe_ports port 70               # gopher
acl Safe_ports port 210              # wais
acl Safe_ports port 1025-65535       # unregistered ports
acl Safe_ports port 280              # http-mgmt
acl Safe_ports port 488              # gss-http
acl Safe_ports port 591              # filemaker
acl Safe_ports port 777              # multiling http
http_access deny !Safe_ports
http_access deny CONNECT !SSL_ports
http_access allow localhost manager
http_access deny manager

#Autoriser le LAN
http_access allow monlan

http_access allow localhost
http_access deny to_localhost
http_access deny to_linklocal
include /etc/squid/conf.d/*.conf
http_access deny all
http_port 3128
coredump_dir /var/spool/squid
refresh_pattern ^ftp: 1440 20% 10080
refresh_pattern -i (/cgi-bin/|\?) 0 0% 0
refresh_pattern . 0 20% 4320

#configuration de base

cache_effective_user proxy
cache_effective_group proxy
cache_mem 16 MB
cache_dir ufs /var/spool/squid 120 16 128
cache_replacement_policy heap LFUDA
cache_swap_low 80
cache_swap_high 90
```

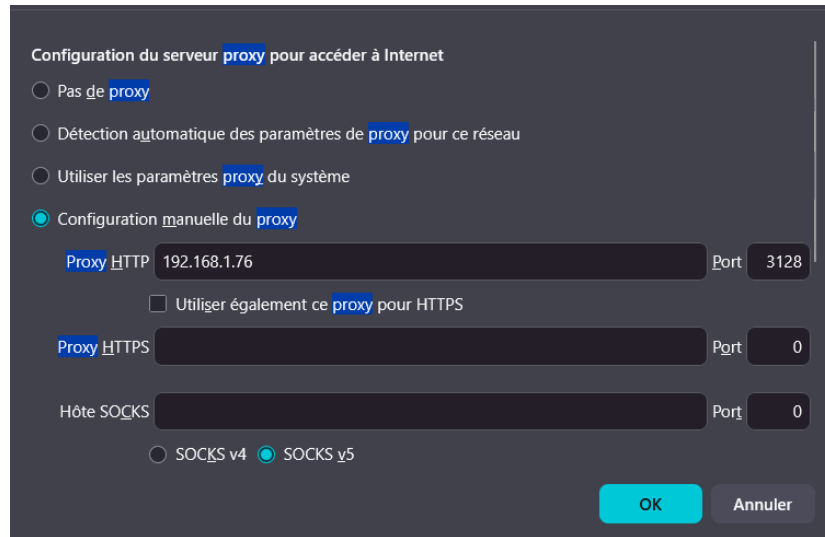
Recharger le service squid

```
# systemctl reload squid
```

Pour tester le proxy utiliser le navigateur Firefox, ou l'on peut configurer le proxy du navigateur.

Allers dans les paramètres du navigateur et chercher proxy

Configurer le proxy de cette façon. Utiliser l'IP de votre serveur



Tester et consulter les logs

```
# tail -f /var/log/squid/access.log
```

```
root@default:~# tail -f /var/log/squid/access.log
1772900070.321    35  :::1 TCP_MISS/301 1063 GET http://www.google.fr/ - HIER_DIRECT/172.217.2
2.99 text/html
1772902442.124    51  192.168.1.115 TCP_TUNNEL/200 6591 CONNECT www.google.com:443 - HIER_DIR
ECT/172.217.22.4 -
1772902471.237    116 192.168.1.115 TCP_TUNNEL/200 4892 CONNECT ads.mozilla.org:443 - HIER_DI
RECT/2a04:4e42:6a::347 -
1772902471.261    140 192.168.1.115 TCP_TUNNEL/200 4914 CONNECT ads.mozilla.org:443 - HIER_DI
RECT/2a04:4e42:6a::347 -
1772902471.339    48  192.168.1.115 TCP_TUNNEL/200 3943 CONNECT img-getpocket.cdn.mozilla.net
:443 - HIER_DIRECT/34.120.237.76 -
```

Les logs remontent bien.

Contrôle horaire :

Ajouter dans /etc/squid/squid.conf

```
# nano /etc/squid/squid.conf
```

A écrire juste avant l'autorisation du lan (http_access allow monlan)

```
acl allowed_hosts src 172.17.13.14
acl limithour time 16:00-17:30
http_access allow allowed_hosts limithour
```

L'heure dans l'acl limithour peut être modifier pour autoriser uniquement le trafic quand vous le souhaitez.

Votre fichier devrait ressembler à a cela.

```
http_access deny manager

#Controle de horaire
acl allowed_hosts src 172.17.13.14
acl limithour time 00:00-23:59
http_access allow allowed_hosts limithour

#Autoriser le LAN
http_access allow monlan
```

Recharger le service squid

```
# systemctl reload squid
```

4. Authentification utilisateurs (NCSA)

Créer fichier utilisateurs

```
# touch /etc/squid/squidusers
```

Ajouter deux utilisateurs avec mot de passe

```
# htpasswd -b /etc/squid/squidusers user1 user1
# htpasswd -b /etc/squid/squidusers user2 user2
```

Modifier squid.conf (au début du fichier)

```
# nano /etc/squid/squid.conf
```

Au début du fichier

```
auth_param basic program /usr/lib/squid/basic_ncsa_auth
/etc/squid/squidusers
auth_param basic children 2
auth_param basic credentialsttl 3 hours
auth_param basic realm Squid proxy SI02A
```

A la fin des ACL

```
acl utilisateurs proxy_auth REQUIRED
```

Au dessus de l'autorisation du lan (http_access allow monlan)

```
http_access allow utilisateurs
```

Votre fichier de conf devrait ressembler a ça.

```
#Authentification utilisateurs
auth_param basic program /usr/lib/squid/basic_ncsa_auth /etc/squid/squidusers
auth_param basic children 2
auth_param basic credentialsttl 3 hours
auth_param basic realm Squid proxy SIO2A

#Autoriser le LAN
acl monlan src 192.168.1.0/24

acl localnet src 0.0.0.1-0.255.255.255 # RFC 1122 "this" network (LAN)
acl localnet src 10.0.0.0/8 # RFC 1918 local private network (LAN)
acl localnet src 100.64.0.0/10 # RFC 6598 shared address space (CGN)
acl localnet src 169.254.0.0/16 # RFC 3927 link-local (directly plugged) machines
acl localnet src 172.16.0.0/12 # RFC 1918 local private network (LAN)
acl localnet src 192.168.0.0/16 # RFC 1918 local private network (LAN)
acl localnet src fc00::/7 # RFC 4193 local private network range
acl localnet src fe80::/10 # RFC 4291 link-local (directly plugged) machines

acl SSL_ports port 443
acl Safe_ports port 80 # http
acl Safe_ports port 21 # ftp
acl Safe_ports port 443 # https
acl Safe_ports port 70 # gopher
acl Safe_ports port 210 # wais
acl Safe_ports port 1025-65535 # unregistered ports
acl Safe_ports port 280 # http-mgmt
acl Safe_ports port 488 # gss-http
acl Safe_ports port 591 # filemaker
acl Safe_ports port 777 # multiling http

#Authentification utilisateurs
acl utilisateurs proxy_auth REQUIRED

http_access deny !Safe_ports
http_access deny CONNECT !SSL_ports
http_access allow localhost manager
http_access deny manager

#Controle de horaire
acl allowed_hosts src 172.17.13.14
acl limithour time 00:00-23:59
http_access allow allowed_hosts limithour

#Authentification utilisateurs
http_access allow utilisateurs

#Autoriser le LAN
http_access allow monlan

http_access allow localhost
http_access deny to localhost
http_access deny to linklocal
include /etc/squid/Conf.d/*.conf
```

Donner les bons droits

```
# chown proxy:shadow /usr/lib/squid/basic_ncsa_auth
# chmod 2750 /usr/lib/squid/basic_ncsa_auth
```

Redémarrer le service squid

```
# systemctl restart squid
```

5. SquidGuard (filtrage avancé)

Installer squidGuard

```
# apt install -y squidguard
```

Télécharger liste noire

```
# wget http://cri.univ-tlse1.fr/blacklists/download/blacklists.tar.gz
```

Décompresser dans le répertoire squidguard

```
# tar -xvzf blacklists.tar.gz -C /var/lib/squidguard/db
```

Ajouter dans /etc/squid/squid.conf

```
# nano /etc/squid/squid.conf
```

A la fin du fichier ajouter les ligne suivant

```
url_rewrite_program /usr/bin/squidGuard -c
/etc/squidguard/squidGuard.conf
url_rewrite_children 10
```

Votre fichier de configuration actuel.

```
#configuration de base
cache_effective_user proxy
cache_effective_group proxy
cache_mem 16 MB
cache_dir ufs /var/spool/squid 120 16 128
cache_replacement_policy heap LFUDA
cache_swap_low 80
cache_swap_high 90

#SquidGuard (filtrage avancé)
url_rewrite_program /usr/bin/squidGuard -c /etc/squidguard/squidGuard.conf
url_rewrite_children 10
```

Créer fichier /etc/squidguard/squidGuard.conf

```
# touch /etc/squidguard/squidGuard.conf
```

```
nano /etc/squidguard/squidGuard.conf
```

Ajoutez à la toute fin

```
dbhome /var/lib/squidguard/db/blacklists
logdir /var/log/squidguard

src lan {
    ip 172.17.1.10-172.17.1.100
}

dest games {
    domainlist games/domains
    urllist games/urls
    redirect http://127.0.0.1/proxy.html
}

acl {
    lan {
        pass !games all
    }
    default {
        pass none
        redirect http://127.0.0.1/proxy.html
    }
}
```


Reconstruire la base

```
# squidGuard -C all
```

Donner les droits à proxy

```
# chown -Rf proxy:proxy /var/lib/squidguard/db/blacklists
```

Créer la page de blocage

```
# echo "<h1>Accès interdit - site bloqué</h1>" >  
/var/www/html/proxy.html
```

Redémarrer squid

```
# systemctl restart squid
```

Vérifier logs

```
# tail -f /var/log/squidguard.log  
# tail -f /var/log/squid/access.log
```

6. Analyseur de log (SARG)

Installer sarg

```
# apt install -y sarg
```

Modifier /etc/sarg/sarg.conf

```
# sed -i 's|^access_log.*|access_log /var/log/squid/access.log| '  
/etc/sarg/sarg.conf  
" sed -i 's|^output_dir .*|output_dir /var/www/html/reports/| '  
/etc/sarg/sarg.conf
```

Générer rapport

```
# sarg -x
```

Créer lien symbolique

```
# ln -s /var/www/html/reports /var/www/html/sarg
```

Consulter via navigateur

[http://\[IPServeurProxy\]/sarg/](http://[IPServeurProxy]/sarg/)

7. Auto-configuration navigateur

Créer fichier PAC

```
cat << 'EOF' > /var/www/html/proxy.pac
function FindProxyForURL(url,host) {
    return "PROXY proxy.sio.local:3128; DIRECT";
}
EOF
```

8. Authentification LDAP

Installer outils LDAP

```
apt install -y ldap-utils
```

Modifier /etc/ldap/ldap.conf (exemple)

```
echo "
BASE    dc=rezo,dc=home
URI      ldap://172.16.3.14
" > /etc/ldap/ldap.conf
```

Modifier /etc/squid/squid.conf

```
echo "
auth_param basic program /usr/lib/squid/basic_ldap_auth -b
ou=posixaccounts,dc=rezo,dc=home -u uid -h 172.16.3.14
" >> /etc/squid/squid.conf
```

Redémarrer squid

```
systemctl restart squid
auth_param basic program /usr/lib/squid/basic_ldap_auth -b
ou=posixaccounts,dc=rezo,dc=home -u uid -h 172.16.3.14
" >> /etc/squid/squid.conf
```